



Retrieval-Augmented Generation for Multilingual Industrial Engineering Documentation: Evaluation and Abstention

Industry Collaboration with Everllence SE (formerly MAN Energy Solutions)

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Abstract

Engineering organisations hold large multilingual corpora of manuals, service reports, and norms; finding the correct procedure or parameter in them is a daily time cost. Retrieval-augmented generation (RAG) [1] enables natural-language access to such corpora. This thesis builds and evaluates a RAG system with Everllence as industrial partner and adds one focused scientific increment: *abstention*. In safety-critical maintenance, a fluent but wrong torque value is worse than no answer, so the system must decline when evidence is insufficient. The work builds a modular pipeline (table-aware ingestion, hybrid retrieval [7, 6], Azure OpenAI + one open-weight generator), constructs an expert-validated evaluation set *including unanswerable questions*, and measures correctness, groundedness (RAGAS/ARES [3, 4]), and the abstention trade-off, validating automatic metrics against expert judgement.

Motivation and Gap

Everllence engineers consult documentation spanning product generations, languages (German/English), and formats [8]; search is keyword-based and retrieval expertise is concentrated in senior staff. Building a RAG demo is easy — knowing whether it can be trusted is not: technical facts live in tables and variant-specific exceptions, hallucination is a documented failure mode [2], and its cost here is asymmetric. Automatic evaluation frameworks [3, 4] and benchmarks measuring “negative rejection” [5] exist, but have been validated mainly on open-domain corpora. Whether their scores track *expert* judgement on dense engineering documentation — and whether prompting-based abstention gives a usable wrong-answer/coverage trade-off — are open, answerable questions.

Feasibility

- Bounded workload: one corpus pair, ~4 retrieval configurations, 2 generators, prompt-level abstention only — the scientific effort concentrates on evaluation design, not risky engineering.
- Mature tooling (open-source ingestion/retrieval frameworks, RAGAS); Everllence’s Azure OpenAI licences remove infrastructure risk.
- Data risk hedged: methodology is developed on a public technical-documentation twin corpus (open engine manuals, classification-society rules); company documents enrich the study but are not blocking.
- Expert effort bounded: two batched workshops (question authoring; blinded rating).

Research Questions

1. **RQ1:** How accurately and groundedly does a well-configured RAG pipeline answer realistic engineering questions over multilingual technical documentation, and which retrieval configuration (chunking; BM25/dense/hybrid) performs best?
2. **RQ2:** Can prompting-based abstention reliably reject unanswerable questions, and at what false-refusal cost on answerable ones?
3. **RQ3:** How well do automatic RAG metrics agree with domain-expert judgement on this corpus — can routine evaluation be automated for regression testing?

Methodology

- **Corpora:** public twin corpus (German/English manuals, rules, standards) for reproducibility + curated non-confidential Everllence set in the company’s Azure tenant; table-preserving, structure-aware ingestion.
- **Evaluation set:** 150–200 QA pairs authored with Everllence engineers across four types (factual, table lookup, procedure synthesis, *unanswerable*), with supporting-passage annotations.
- **Pipeline:** interchangeable chunkers and retrievers (BM25, multilingual dense, hybrid; optional reranker); generators: one Azure-hosted and one open-weight model, identical prompts; no fine-tuning.
- **Abstention:** prompt-level refusal with evidence-sufficiency self-check; measured as rejection rate (unanswerable) vs. false-refusal rate (answerable).
- **Evaluation:** retrieval recall@k/nDCG; answer correctness (expert rubric), groundedness (RAGAS [3]), citation accuracy; metric validation via agreement statistics between automatic scores and blinded expert ratings (~200



answers, 2–3 raters).

- **Scope/risk:** no agentic retrieval, no diagram understanding, no deployment. All RQs answerable on the public corpus alone if company access is delayed; LLM-judge fallback if expert capacity shrinks.

Contribution and Value for Everllence

(1) A quantitatively evaluated multilingual RAG assistant for industrial documentation; (2) an expert-validated measurement of abstention quality (negative rejection vs. false refusal) on real technical documents; (3) a released bilingual evaluation set and pipeline enabling regression testing. For Everllence: a deployment decision basis — measured accuracy, groundedness, and refusal behaviour on its own document types and Azure infrastructure.

Timeline

Month	Work
1	Literature review; twin-corpus collection; table-aware ingestion; question typology.
2	Evaluation-set authoring (workshop 1); baseline pipeline end-to-end.
3	Retrieval experiments (RQ1); open-weight generator comparison.
4	Abstention implementation and evaluation (RQ2); error analysis by type and language.
5	RAGAS runs; blinded expert rating (workshop 2); metric validation (RQ3); company-corpus case study.
6	Thesis writing, revision, release, submission.

References

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